Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/14/2011
PI Name:	Haley, Gwendolen Ph.D.		
Project Title:	Effects of Alphalipoic Acid on Irradiation Induced Cognitive Deficits		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Oregon Health & Science University		
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City:	Portland	**State:**	OR
Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 NSBRI-RFA-10-01 Postdoctoral Fellowships
Start Date:	11/01/2010	**End Date:**	10/01/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project ended early per NSBRI reports. Original end date was 10/31/2012 (Ed., 11/17/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Raber, Jacob (MENTOR/Oregon Health & Science University)		
Grant/Contract No.:	NCC 9-58-PF02301		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP The Specific Aims of this project are: Aim 1. To determine the short-term effects of 56Fe radiation irradiation on hippocampal function and assess whether these effects are sex-dependent. Mice will be irradiated at 0, 0.1, 0.2, or 0.5 Gy and behaviorally tested 1-2 weeks following radiation. Aim 2. To determine whether the severity of the cognitive effects are associated with reduced ChAT immunoreactive cell bodies in the medial septum and nucleus basalis and ChAT immunoreactive fibers in the hippocampus and cortex. Following behavioral testing, ChAT cell number and fiber density will be determined using immunohistochemistry and		

Task Description:	measures of ROS by western blot. Aim 3. To determine the ability of the dietary supplement LA to antagonize the effects of 56Fe irradiation on hippocampus-dependent cognitive function, measures of ROS, and ChAT immunoreactive cell bodies in the medial septum and nucleus basalis and ChAT immunoreactive fibers in the hippocampus and cortex. Mice will be sham-irradiated or irradiated at a dose causing maximal cognitive injury in Aim 1A and receive regular diet or LA-containing diet starting 1 week prior to behavioral testing. Brains will be processed as in Aim 2. Thus far, we have found that there are early effects of 56Fe irradiation on cognitive function, and that these are sex-dependent. Low doses of irradiation, had differential effects on cognitive function in three distinct hippocampus-dependent cognitive tasks. Compared to sham-irradiation, radiation impaired, in a dose-independent fashion, novel object recognition in female and male mice and spatial memory retention in the first water maze probe trial in male mice. In the second water maze probe trial, sham-irradiated and 0.2 Gy-irradiated male mice showed spatial memory retention but none of the other groups did. In the third water maze probe trial, sham- and 0.2 Gy-irradiation male mice and sham- and 0.5 Gy-irradiated female mice showed spatial memory retention but none of the other groups did. In contrast to novel object and spatial memory retention in the water maze, there were no effects of irradiation on contextual fear conditioning. Following cognitive testing, the mice were sacrificed and their brains were quickly removed. The hippocampus and the cortex were dissected and analyzed for measures of oxidative stress (3-nitrotyrosine) and acetylcholine function (muscarinic type 1 (m1) receptor and acetylcholinetransferase (ChAT)). For hippocampal but not cortical, 3NT and ChAT levels, there were sex x treatment interactions. Compared to sex-matched sham-irradiated mice, irradiated male mice had higher but irradiated female mice lower levels of 3NT and ChAT. There were no group differences in hippocampal or cortical M1 levels. Therefore, we conclude that while hippocampus-dependent cognitive performance and hippocampal 3NT and ChAT levels are sensitive to early effects of 56Fe irradiation, there are no simple relationships between them. These findings support our hypothesis that there are short-term effects of 56Fe irradiation on hippocampal function and that these effects are sex- and task-dependent. Furthermore, it seems that 3NT and ChAT are differentially affected by irradiation in a sex-dependent fashion. These findings complete the first two aims of this proposal and a manuscript with these data has been submitted for publication. This project has an end date of 09/30/2011 as Dr. Haley will be starting a new position. Starting September 2011, Dr. Raber and Lauren Yeiser, a research assistant in the Raber Laboratory, will assess the ability of alpha-lipoic acid to anatagonize the early 56Fe radiation-induced cognitive impairments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the mechanisms of irradiation damage and what tissues are susceptible can have lasting impact not only in space radiation, but also for radiation exposure as part of flights, cancer treatments, and potential 'dirty bomb' scenarios.
Task Progress:	During the last year, the first two aims outlined for this project were completed. First, we determined short-term effects of 56Fe irradiation on spatial learning and memory, novel object recognition and fear conditioning in C57Bl6J male and female mice. 56Fe irradiation had differential effects on cognitive performance in a dose, but not sex, dependent manner. Moreover, these deficits were associated with sex-dependent changes in the reactive oxygen species (ROS) marker 3NT and in the acetylcholine marker ChAT. This work was recently submitted for publication and will be presented at the Society for Neuroscience conference in November of 2011. Starting in September 2011, Dr. Raber and Lauren Yeiser, a research assistant in the Raber Laboratory, will assess the ability of alpha-lipoic acid to antagonize early 56Fe radiation-induced cognitive impairments. Dr. Haley's project will end 9/30/2011 as she will be moving on to a new position.
Bibliography Type:	Description: (Last Updated: 11/17/2014)
Articles in Peer-reviewed Journals	Haley GE, Villasana L, Dayger C, Davis MJ, Raber J. "Apolipoprotein E genotype-dependent paradoxical short-term effects of (56)fe irradiation on the brain." Int J Radiat Oncol Biol Phys. 2012 Nov 1;84(3):793-9. Epub 2012 Mar 6. http://dx.doi.org/10.1016/j.ijrobp.2011.12.049 ; PubMed PMID: 22401921 , Nov-2012
Articles in Peer-reviewed Journals	Haley GE, Yeiser L, Olsen RH, Davis MJ, Johnson LA, Raber J. "Early effects of whole-body (56)Fe irradiation on hippocampal function in C57BL/6J mice." Radiat Res. 2013 May;179(5):590-6. http://dx.doi.org/10.1667/RR2946.1 ; PubMed PMID: 23510274; PubMed Central PMCID: PMC4117374 , May-2013
Awards	Haley GE. "Medical Research Foundation Grant, March 2011." Mar-2011